Master Advisory Digital Crop Survey (DCS)

Subject: State Readiness Matrix for taking up Crop Survey Project during Kharif-2023.

The Crop Survey Project is an ambitious initiative of the Union Government and the 12 State Governments that are to be covered during Kharif – 2023. Gradually, all other states would be onboarded into the Crop Survey Project. The uniqueness of this project, as far as execution is concerned, is from the fact that crop survey is a time-limited operation and would have to be completed within about 45 days from the day of initiation. Generally, Crop Survey would start in a tehsil/taluka or a village one month after the crop is sown and should be completed about 15 days before the harvest. Taking a normal crop duration of three months, there will be a 45-day window for the crop survey to be finished for all the plots.

As the quantum of work is large with all the crops of all the farmers in each of their plots to be surveyed with verifiable photographs, it is expected that for these 11 states anywhere between 1.5 – 2.0 lakh crop surveyors would be involved. Hence, it is important that all preparations are done by the State and the Central Government. While the crop survey work will be all done by the State Government, it is the responsibility of the Centre to support the state initiative in all possible manners.

Central Government's Responsibilities

1. The Crop Survey Mobile and Web App

While the State Government is free to use their apps after incorporating the mandatory features indicated in the ministry letter dt 06/04/2023 from Additional Secretary, those states who want to use the central app are most welcome to do so. It would be for the state to configure the central app on some parameters like the start and end of the season, the Taluka names, etc. A detailed list of all the parameters to be configured would be sent separately. The State Government would also have to connect/link a copy of their land records and geo-reference maps to the Crop Survey Application. The complete documentation would be provided by the Centre along with support for connecting the two databases to the application.

2. The User Manual

It would be the Central Government's responsibility to provide a user manual in English. However, the translation into the local/official language would have to be done by the State Government.

3. Training Documents

As the crop survey would involve a large number of Crop Surveyors, Supervisors, Verifiers, etc., it is extremely important that training documents are prepared and supplied to all the concerned. The Central Government will prepare videos, PPTs, and training manuals for this purpose. However, these would be prepared in English and would have to be translated into the local/official language by the State Government.

4. Training App

The Central Government is also creating a training mobile app that can be used by all including Crop Surveyors, Supervisors, and Trainers to self-train themselves through the training app. The training app would have ALL the features of the actual Crop Survey App ensuring the familiarity and comfortability of the surveyors with the application and its functionalities having worked and practiced with sample data at their home or in training centers.

5. Video-Based User Manual

The Central Government would also prepare a video manual with separate training for each of the players namely Crop Surveyors, Supervisors, Tehsil/Taluka Officers, and state-level officers. This would make it convenient for the officials to easily understand the working of the entire system. Various videos on domain (small tutorial videos of 3-5 minutes) would be prepared in English explaining the whole philosophy, the need for and importance of crop survey, the definition/description of land records, and the implementation of survey operations. These videos will be created assuming no previous knowledge of the matter and are expected to make the users aware and well-versed in the domain of land records and agriculture ensuring proper learning and training for all.

6. Setting up of Help Desk

Each State Government may like to set up a help desk that can respond to calls from the crop surveyors, supervisor, and master trainers in case they face any technical or understanding problems in the field while carrying out a crop survey or uploading the data (huge amount of data to be uploaded every day after daily operations have closed). The help desk would be an extremely important initiative. The Central Government would suggest/assist as to how the help desk can be set up although the state would be free to set up the same in their own way.

7. Technical Support Team

While the required system, software, and hardware, will have to be provided in the data centre or in private or public cloud by the State Government, it would be important to provision for technical management/handling of the project. The Central Government would suggest what type of manpower may be required for handling such technical work. Again, State

Government would have the freedom to set up the technical support team in whatever way they feel would be most appropriate.

8. Master Trainers

The role of the Master Trainer is very highly critical and essential to the project. They are supposed to train the District Level Trainers crop surveyors, supervisors, etc., and help them by solving their problems as and when any arise in those 45 days. Central Government would provide guidance on the type and number of master trainers to be accounted for along with their roles and responsibilities. Representatives of the Central Government would visit various states to help the administration in the training of the master trainers in the state capital.

9. District Trainers

The role of the District Trainer is also very critical and essential to the project. They are supposed to be the single point of contact for guiding and supporting crop surveyors, supervisors etc, and help them by solving their problems as and when any arise in those 45 days. This is because the surveyors would not be aware of the crop survey nuances. Each District Trainer will have not more than 20 surveyors under them as otherwise it would just be impossible to handle the work effectively.

10. Roles and Responsibilities of each Stakeholder

The project has various stakeholders including the State Chief Secretary, State Development Commissioner, State Revenue Secretary, State Agriculture Secretary, Collector/Deputy Commissioners, Tehsildars, Agriculture officers, Taluka Agriculture Officer, Village Agriculture Officer, Village Accountants/Patwari etc. The Central Government will provide a draft of recommended/suggested roles and responsibilities of each stakeholder for their guidance.

State Government's Responsibilities

The followings are the items on which the states will need to act to be ready for the successful execution of the project. In that sense, it can be referred to as a Readiness Matrix for the State Government. The Readiness Matrix can be divided into two parts:

1. Preparatory Work for IT System
2. Manpower

Preparatory work for IT System

1. Required hardware and software to be provisioned in State Data Centre/Cloud.

Crop survey software would require good quality hardware and software systems such as an Operating System and Database. The configuration of the hardware and the software has already been indicated in the Ministry's letter no. Z-11021/60/2022-Digital dated 11th May 2023. This infrastructure will have to be immediately provided and managed properly. It may be noted as per the experience of various states that when the data gets uploaded in the evening by almost all the crop surveyors together, there would be peak load on the system, and accordingly backend will need to dynamically provision for and manage more memory, CPU or when/if needed, more servers. The state will have to request its IT department to keep these system requirements in mind.

2. Technical Support to Crop Survey

As has been mentioned before, there would have to be a technical team to support the crop survey operation. This would typically include the database administrator, system administrator, and others.

3. State-level PMU

The State-level PMU would be a combination of administrative and technical staff. The administrative staff would be from the agriculture, revenue, and horticulture department with the required level of competence and a keen interest in such technical work. The State-level PMU would also have technical staff that will need to be hired from the market or arranged from the IT department. The technical staff would be responsible for handling the Data Base Administrator (DBA) work and the System Administrator Work. They would also help the software team to understand any bugs and take remedial measures. They would coordinate with the help desk team to debug the software from time to time and report remedial action to be taken to the software team.

4. Data Preparation

The two important datasets required for crop survey to work are Land Record Data (ROR) and the Geo-reference Village Maps. The required structure of the ROR data and the village maps data has already been indicated to you in the ministry's letter no. Z-11021/1/2022-DA dated 12.04.2023. The data would have to be extracted from the Land Records Database and made available to the crop survey system right before the crop survey operations start. It must be kept in mind that if data is captured earlier, it may become stale as some mutation may have taken place. If considered appropriate, the land records data can also come through API from the land record system. This would be for State Government to decide how they want to provide Land Records Data to the Crop Survey Application. However, the Agri

Stack Software Team would support the state's technical team in this item of work. As the national application would work strictly as per the LGD code, the data will need to either mandatorily have the village code as per the LGD code or the village codes will need to be mapped as per the LGD code.

5. Village map Shape Files and Index Data

As explained in the previous point, the state government will need to provide, to its crop survey application, connectivity to the required shape files and index data for geo-reference maps.

6. IT System Architecture

The states may decide whether they would like to provide separate Web Servers and Database Servers for various divisions in the states to manage the load on their system. Their Technical Architecture Team would be competent to handle this aspect. Again, the Agri Stack Software Team and NIC Karnataka Team (responsible for writing the software) would be able to suggest various permutations and combinations in this regard.

7. Various Data Masters

There would be various master data registries required for undertaking crop survey. They would typically include the Crop Master, the Irrigation Sources Master, the Land Use Master, etc. A separate circular of master data registries would be issued in due course.

8. District Taluka Village Master

The state will have to finalize these masters in accordance with the LGD Code.

9. Translating the Training Material, User Manual, and Videos into Local/Official language

As has been already mentioned, all the material would be provided in English by the Central Government and the State Government will need to translate them into the local/official language. They would also have the liberty to modify the same if so required.

10. Carrying Out User Acceptance Test (UAT)

While the UAT would be done as part of POC by each state, a real UAT would mean that the app is tried in the actual environment of hardware and software installed at the state level. It would be extremely important that a UAT is done in the real environment before the project starts so that any unexpected problem caused by the installation of the hardware and software can be found and taken care of.

Manpower and Training

1. State PMU

While the role of State-level PMU has already been clarified above in the document, it is important to understand its criticality from the point of view of driving this project. The success of crop survey depends upon the agility and competence of the State-level PMU. States should very carefully choose the required officers for the PMU and the technical staff and empower them to review the project. It may be kept in mind that as the crop survey would involve officials from agriculture, horticulture, and revenue, the State PMU will need to be accordingly constituted.

2. Various Committees

The State Level Committee chaired by Development Commissioners, the District Level Committee under the chairmanship of the Collector, and a Taluka Level Committee under the chairmanship of Tehsildars would have to be constituted for monitoring the project and taking various decisions. While all the policy decisions should be taken at the state level committee, the district level and taluka level committee would also be very important as they will take various implementational level decisions and would continuously monitor the project which has to get completed within a period of 45 days.

3. Provisioning of crop surveyor resources

Crop survey requires large manpower. The government will have to decide whether they want to do the survey using government staff or the local youth from the same village. As per the scheme guidelines, only these two types of resources are permitted. NO company or organization can be hired to provide resources on a contractual basis. The resources must be from the local village as local knowledge and belongingness to the project along with accountability to local farmers is essential. It would require efforts to find and finalize crop surveyors in case the state decides to hire local youth for the same.

Similar efforts would be required even if crop survey is to be done using government manpower. It must be kept in mind that for those 45 days, these government resources might need to work exclusively on this project and will need to be instructed and empowered to take up this crop survey work even at the cost of other duties for a short period.

Further, based on knowledge, competence, and belongingness to the project, a rational mix of government officers from the agriculture, horticulture, and revenue department would have to be selected for the crop survey. As the crop survey would start shortly, it is of utmost importance that their selection is made immediately and not later than the end of the month.

The number of surveyors would depend upon the number of owner plots (if the survey is done owner plot-wise) or the number of survey numbers (if the survey is done survey-number-wise). It should be ensured that no crop surveyor is assigned more than 1500 owner plots or survey numbers, as the case may be, in a season. There would be a strong temptation to assign a greater number of owner plots/survey numbers to a surveyor but the experience in other states has shown that the quality of the survey is compromised in this scenario. It may also be noted that if a survey is not done in time the crop may get harvested and the crop surveyor would not be able to conduct the crop survey.

In case the crop surveyors are local youth, the government will have to decide as to which official would be responsible to finalize the crop surveyors for each village, i.e., the Village Accountant/Patwari or the agriculture or horticulture staff.

4. Master Trainers

The importance of master trainers cannot be emphasized more. They are the key players in successfully conducting crop surveys. The crop surveyors, whether government officials or private workers, would be unaware of the crop survey nuances. They will have to be handheld and trained. This would be the work of the master trainers. It is suggested that there should be at least 4-5 master trainers in each Taluka/Tehsil and that one master trainer should not handle more than 20 crop surveyors as otherwise it would just be impossible to handle the work effectively.

5. Roles and Responsibilities of Stakeholders

The State Government will have to issue clear orders fixing the roles and responsibilities of each committee and each stakeholder. This would help in avoiding any ambiguity in the work and diffusion of responsibility.

6. Training of the Master Trainers

While the central ministry, along with their experts, would go to each state and train the master trainers for one day, it is expected that the resource people of the state government would, once or twice again, train the master trainers. Each master trainer in turn would have to train the crop surveyors.

7. Training of Crop Surveyors

A schedule would have to be fixed under the supervision and accountability of the Tehsildar to see that all crop surveyors are trained by master trainers in an intensive manner using the same training material through which they were trained. Enough physical and/or electronic copies of these training materials will have to be created by the state government. Please

note that training of master trainers and crop surveyors must be done physically and not through video links as that hamper the effectiveness of the said training.

During the training of the master trainers and the crop surveyors, various exercises would have to be given to them so that they can try the same on the training app.